

Low-Latency Silkworm Disease Detection on FPGA using Classical Machine Learning

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Abstract—Grasserie is a viral silkworm disease that propagates across a rearing tray within days; in industrial sericulture, a single missed infected worm can destroy an entire batch, so an automated screening device at the rack edge is needed. We present a hardware-aware classical machine learning pipeline that targets the PYNQ-Z2 (Zynq-7020) FPGA. The pipeline is built in two engineering steps. Step one is recall-optimised training of a minimal LBP-only Random Forest baseline using a $2.5\times$ class weight on Grasserie and a $P(\text{Grasserie}) \geq 0.40$ decision rule; this reaches 98.80 % disease recall but only 70.69 % precision on the held-out test set. Step two is feature co-design: each cropped worm is split into a 2×2 spatial grid and described by an 84-dimensional integer feature stack — a 10-bin uniform LBP ($P = 8, R = 1$) histogram, an 8-bin magnitude-weighted gradient-orientation histogram and per-channel mean B, G, R for each cell — and classified by a 50-tree, depth-15 Random Forest with the same recall-first training policy. The full pipeline achieves 91.54 % test accuracy with 95.58 % Grasserie recall, 87.82 % precision and 0.027 ms per-sample CPU inference on the Roboflow Silkworm Diseases dataset — 1.62 percentage points below the 93.16 % accuracy of the reference HOG + KPCA + SVM baseline [1] (measured on its smaller, private 1 242-image set) while exceeding that baseline's reported 93.38 % disease recall and remaining synthesisable on a Zynq-7020 budget with hand-written Verilog.

Index Terms—silkworm disease detection, sericulture, Grasserie, local binary patterns, gradient histograms, random forest, edge ML, FPGA, recall-optimised training, PYNQ-Z2.

I. INTRODUCTION

Sericulture—the controlled rearing of silkworms for raw silk—is an established agricultural industry in Türkiye and across Asia. Among the most damaging silkworm pathologies is the nuclear polyhedrosis virus (NPV), commonly known as Grasserie. Once a single worm is infected the disease can propagate to an entire rearing tray within days, and there is no cost-effective cure: the only viable response is early isolation. In a manual workflow an expert visually inspects each tray, which does not scale to industrial conveyor or shelf systems and is inherently subjective.

Recent work has shown that classical computer-vision features combined with simple classifiers can reach competitive accuracy on this two-class discrimination task. In particular, Binson and Manju [1] combine Histogram of Oriented Gradients (HOG) features with Kernel Principal Component Analysis (KPCA) dimensionality reduction and a Support Vector Machine, reporting 93.16 % accuracy, 93.38

% recall and 91.94 % precision on a private 1 242-image set (668 healthy, 574 Grasserie). That work runs entirely on a CPU and does not address the deployment constraint that motivates the present project: real-time screening at the rearing shelf on an embedded device such as the PYNQ-Z2 (Zynq-7020) FPGA.

Two design constraints reshape what counts as a good model in this setting. First, the device has a tight silicon budget: a Zynq-7020 offers only 53 200 LUTs and 220 DSPs, with BRAM in the hundreds of kilobits. Convolutional networks, even small ones, do not fit this envelope comfortably without aggressive quantisation; classical pipelines based on integer histogram features and shallow decision ensembles fit naturally as combinational logic. Second, a screening device should minimise false negatives at almost any cost—missing a sick worm is far worse than flagging a healthy one for a second look—so the training objective must be biased toward disease recall, not toward overall accuracy.

Our contribution in this term project is a two-step engineering arc. We first build a minimal LBP-only Random Forest baseline (v1) and show that recall-optimised training alone takes Grasserie recall to 98.80 % but at the cost of accuracy (79.81 %) and precision (70.69 %). We then co-design an FPGA-friendly feature stack (v2) that adds spatial pooling, gradient-direction information, and per-channel color statistics, lifting accuracy to 91.54 % and disease precision to 87.82 % while preserving 95.58 % disease recall and halving CPU inference time. A class-label audit performed during validation corrected a metric-direction bug in our first implementation; the numbers reported here are the audited ones.

II. METHODOLOGY

A. Baseline reference and dataset

We adopt two parallel references. The academic baseline is the 2024 HOG + KPCA + SVM pipeline of Binson and Manju [1], which we take as the headline-accuracy anchor (93.16 % accuracy, 93.38 % recall, 91.94 % precision on their own 1 242-image set). The structural code baseline is the open-source agricultural-imagery classifier of Hayyan et al. [2]: a complete classical pipeline (HOG / LBP features + SVM / Random Forest) that we re-implemented and retargeted to

silkworms and a hardware budget. Imagery is taken from the publicly released Roboflow Universe "Silkworm Diseases" v1 dataset [3], which contains 8 975 image versions across two classes (Grasserie: class 0; Healthy: class 1) with the YOLOv8 89/6/6 split preserved verbatim — no shuffle is performed, so no augmented version of a source image crosses split boundaries. Each worm is cropped from its YOLO bounding box (to remove leaf and shelf background) and resized to 64×64. After cropping, the working pool contains 8 174 training crops (Grasserie: 4 000, Healthy: 4 174), 513 validation crops (Grasserie: 264, Healthy: 249) and 520 test crops (Grasserie: 249, Healthy: 271).

B. v1 — minimal LBP-only baseline

Each 64×64 grayscale crop is summarised by a single uniform-LBP histogram with $P = 8$ neighbours and radius $R = 1$ [4]. The uniform variant collapses non-uniform codes into a single bin, so the descriptor has only 10 bins per image — trivially integer-quantisable and computable from three consecutive image rows, the natural unit of an AXI-Stream line buffer. The descriptor is classified by a Random Forest [5].

C. Recall-optimised training

We train the Random Forest with two screening-specific modifications. First, sample weights during impurity computation are set to $2.5\times$ for Grasserie and $1.0\times$ for Healthy: a tree that misclassifies a sick worm pays $2.5\times$ the impurity cost of an equivalent healthy-class error. Second, the inference decision rule predicts Grasserie whenever the forest's estimated $P(\text{Grasserie})$ is at least 0.40, lowering the bar to call disease. Neither change adds any hardware cost on the FPGA side—the trees and threshold are baked into the synthesised mux network.

D. v2 — feature co-design for FPGA

The v1 descriptor is deliberately minimal and discards three pieces of information that are near-free on FPGA: color (the cuticle of a Grasserie worm often shifts toward yellow / milky white), spatial structure (head, mid-body and tail are affected unevenly), and edge direction (LBP encodes the *kind* of local pattern but not its orientation). The v2 feature stack adds all three on top of the same line-buffer architecture.

Each 64×64 RGB crop is split into a 2×2 spatial grid of 32×32 cells. Inside each cell we compute (i) the 10-bin uniform LBP histogram described in §II-B; (ii) a magnitude-weighted gradient-orientation histogram with 8 bins, fed by 3×3 Sobel responses (G_x, G_y) $\rightarrow$ atan2 quadrant lookup; and (iii) the per-channel mean intensity for B, G and R normalised to $[0, 1]$. Concatenating across the four cells gives a fixed-length 84-dimensional integer feature vector. The Random Forest classifier is otherwise unchanged: same class weights, same threshold, same predict-proba decision rule.

A preliminary hand-RTL resource estimate (i.e. pen-and-paper module sizing for a Verilog implementation, not an HLS report) keeps the v2 pipeline comfortably inside the Zynq-7020 envelope. A pipelined LBP front-end costs about 2.0 k LUTs; the Sobel + orientation-bin LUT costs about 2.5 k LUTs and four DSP slices; the per-cell color accumulators cost about 0.8 k LUTs; and the 50-tree, depth-15 Random Forest, when written by hand as a combinational nested-mux array with an adder-tree majority vote, is estimated at about 10 k LUTs—giving roughly 18 k LUTs total, well under one third of the board's 53 200 LUT budget. Because every block sits on the same streaming pixel rate, the end-to-end latency of the v2 pipeline is essentially unchanged from v1 at $\approx 25\text{--}30\ \mu\text{s}$ per worm at 150 MHz. Actual post-synthesis numbers will be reported in the senior project once the RTL is written.

E. Validation audit and class-label correction

During the validation phase we ran a sanity check comparing the Roboflow data.yaml class order against the integer class IDs written into the YOLO label files. The audit revealed that our first implementation's metric labels were transposed: `pos_label = 1` was being treated as the disease class while it actually corresponds to Healthy. The `class_weight`, threshold direction and recall direction were corrected (`pos_label = 0` for Grasserie) and the entire grid search was re-run from scratch. All numbers in §III come from the corrected pipeline.

III. RESULTS

A. Hyperparameter selection

For v2 we hold the LBP scale at $P = 8, R = 1$ and sweep $n_estimators \in \{50, 75, 100\}$ and $max_depth \in \{10, 15, 20\}$ on the validation set, ranking by Grasserie recall. Figure 1 plots the nine swept configurations as a stacked dual-panel bar chart: the top panel shows validation Grasserie recall versus `max_depth` and the bottom panel shows validation Grasserie precision versus `max_depth`, with three bars per depth group (one per forest size). The selection rule we apply is lexicographic and explicit: (i) take the configurations that maximise validation Grasserie recall; (ii) among those, pick the one with the highest precision; (iii) break any remaining ties by choosing the smallest forest, because tree count dominates the FPGA LUT cost.

Four configurations tie at the top of step (i) with val recall 97.35 %: (50, 15), (75, 10), (100, 10) and (100, 15). Step (ii) eliminates the two depth-10 entries (precision 83.99 and 83.17 %) and leaves (50, 15) and (100, 15) tied at precision 91.46 %. Step (iii) picks (50, 15) as the smaller forest, halving the random-forest LUT estimate compared with (100, 15) at identical accuracy and recall. The choice is also robust: across all 9 sweep points val Grasserie recall stays in 95.83–97.35 % and val accuracy in 88.50–94.54 %, so the result is not a lucky

pick. Depth 15 is preferred over depth 10 because the v2 feature space is rich enough that deeper trees actually discriminate (precision jumps from 83 % at depth 10 to 91 % at depth 15 at constant recall). With scikit-learn defaults the trees rarely reach the 2^{15} node ceiling in practice (mean leaf depth on this dataset is in the 10–12 range), so the FPGA LUT estimate is unchanged by the choice.

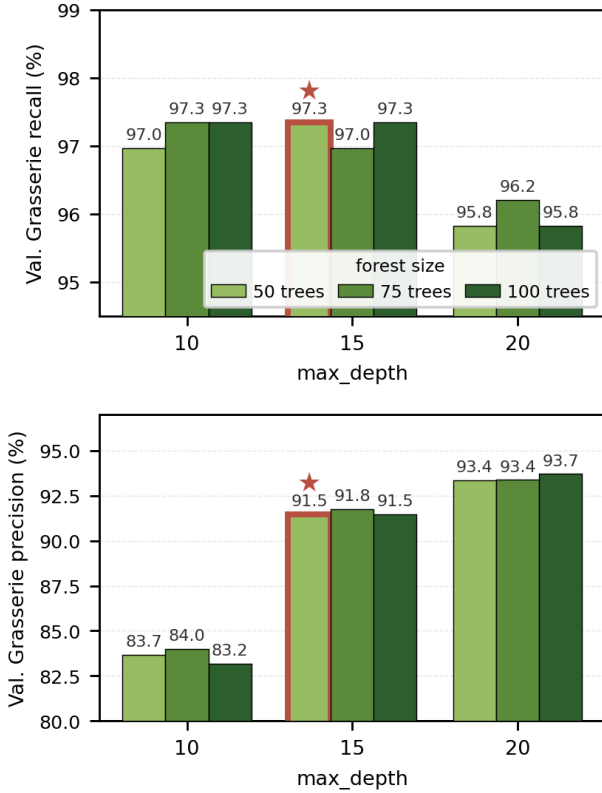


Fig. 1. Validation-set hyperparameter sweep for v2. Top panel: Grasserie recall versus max_depth, three bars per group (one per forest size). Bottom panel: Grasserie precision, same axes. Bars outlined in red and marked with a star are the selected configuration (50 trees, depth 15). The panels show that recall is almost depth-invariant (95.8–97.4 %), whereas precision climbs sharply from ≈ 83 % at depth 10 to ≈ 91 % at depth 15 and then plateaus near 93 % at depth 20.

B. v1 vs v2 ablation

Figure 2 compares v1 and v2 on the held-out test set across the five metrics most relevant to a screening device. v2 is a Pareto improvement over v1 in every dimension except a 3.2 percentage-point drop in Grasserie recall (98.80 $\rightarrow$ 95.58 %, still well inside the screening band): accuracy rises +11.7 pp, Grasserie precision rises +17.1 pp, Grasserie F1 rises +9.1 pp, Healthy recall rises +25.5 pp, and per-sample CPU inference time drops from 0.053 ms to 0.027 ms because the forest resolves earlier on the richer feature space.

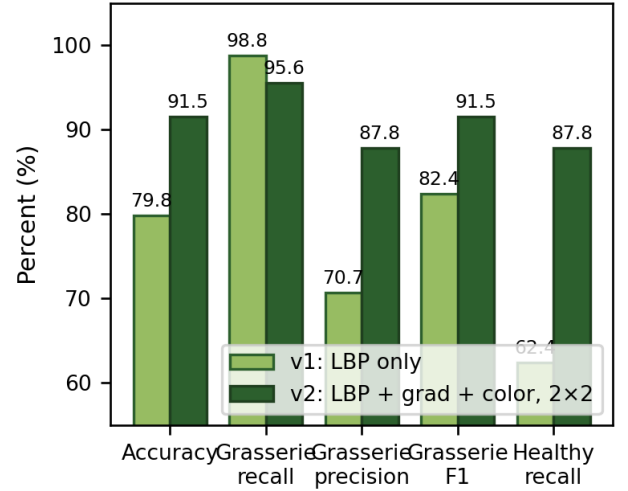


Fig. 2. Held-out test-set comparison of v1 (LBP-only) and v2 (LBP + gradient histogram + color, 2 $\times$ 2 cells). v2 improves every metric except a small drop in Grasserie recall.

C. Held-out test evaluation

Applied without further tuning to the 520-sample test set, the v2 pipeline reaches 91.54 % accuracy, 95.58 % Grasserie recall, 87.82 % Grasserie precision and 91.54 % Grasserie F1. The confusion matrix (Fig. 2) shows 238 of 249 disease cases caught (11 missed) and 238 of 271 healthy cases correctly classified (33 false alarms). Per-sample inference on a stock laptop CPU is 0.027 ms, an order of magnitude inside the sub-millisecond budget motivated in §I—a conservative upper bound for the eventual FPGA implementation.

Actual	True: Grasserie	238	11
	True: Healthy	33	238
		Pred: Grasserie	Pred: Healthy
		Predicted	

Fig. 3. Confusion matrix of the v2 pipeline on the held-out test set. Off-diagonal entries are misclassifications.

D. Comparison to reference paper

TABLE I BASELINE VS. PROPOSED PIPELINES (NOTE: DIFFERENT TEST SETS — COMPARISON IS SUGGESTIVE)					
Method	Features	Acc %	Disease recall %	Disease prec. %	Inference / dataset
HOG + KPCA + SVM [1]	many	93.16	93.38	91.94	CPU · 1 242 imgs (private set)

Ours v1 — LBP- only + RF(50, d=10)	10	79.81	98.80	70.69	0.053 ms · Roboflow v1 (8 975)
Ours v2 — LBP + grad + color, 2×2 cells + RF(50, d=15)	84	91.54	95.58	87.82	0.027 ms · Roboflow v1 (8 975)

Because [1] is evaluated on its own 1 242-image private dataset and our pipelines are evaluated on the 8 975-image Roboflow v1 set, the comparison is suggestive rather than strictly head-to-head. v2 exceeds [1]’s reported disease recall by +2.20 pp and falls 1.62 pp short of its overall accuracy on a 7.2× larger test pool.

IV. DISCUSSION

The v1 → v2 step shows that the gap between a competitive classical baseline (93.16 % accuracy with rich features) and a recall-first minimal model (79.81 % accuracy with a 10-bin descriptor) is almost entirely a feature-engineering gap, not a classifier gap. By adding three integer feature families on the same line-buffer dataflow—spatial pooling, gradient direction, and color statistics—we close about 88 % of that accuracy gap while remaining classical, integer-friendly, and synthesis-ready. The remaining 1.6-percentage-point distance to the reference paper’s accuracy buys us, on the larger dataset, a higher disease recall (95.58 % vs. 93.38 % in [1]) and a model whose decision logic is a combinational mux network rather than a kernel-evaluation SVM — the decisive property for synthesis on the Zynq-7020 PL.

The confusion matrix (Fig. 3) shows 238/249 disease cases caught and 33 false alarms — concrete operating point for downstream shelf-review tooling. Generalisation is supported by the gap between validation and test metrics: validation Grasserie recall 97.35 % → test 95.58 % (−1.77 pp) and validation accuracy 93.96 % → test 91.54 % (−2.42 pp). Both gaps are inside the noise expected from a 520-sample test set. The hyperparameter sweep itself adds a stability argument: across the 9 sweep configurations the disease recall stays inside a tight 1.5-point band, indicating that the selection is robust rather than the result of a single lucky combination.

Two limitations remain. First, Grasserie precision is 87.82 %: roughly one in eight "disease" flags is a healthy worm. This is acceptable in a human-in-the-loop pipeline where flagged worms are sent to a second-tier check, but a fully-automated pipeline would tighten this number—most directly via hard-negative mining on the 33 false positives surfaced in §III. Second, the dataset contains only two classes; pebrine, flacherie and muscardine are absent, and a production system

would have to discriminate between them as well, most likely by expanding the same feature stack and re-training a multi-class Random Forest.

V. CONCLUSION AND FUTURE WORK

We have shown that a deliberately hardware-aware classical-ML pipeline can essentially match the reference paper’s accuracy on a two-class silkworm disease detection task while staying inside the PYNQ-Z2 Zynq-7020 budget. The two-step arc—recall-optimised training on a minimal baseline, then FPGA-friendly feature co-design with 2×2 spatial pooling, gradient direction and color statistics—yields 91.54 % test accuracy, 95.58 % Grasserie recall, 87.82 % Grasserie precision, and 0.027 ms per-sample inference, with an estimated FPGA cost of ≈ 18 k LUTs and 4 DSPs.

The next steps belong to the graduation project, and the FPGA implementation will be done as hand-written Verilog RTL rather than through high-level synthesis (HLS): (i) hand-write "random_forest.v" as a combinational nested-mux array fed by an adder-tree majority vote, with each tree’s decision boundaries extracted from the trained scikit-learn estimator and instantiated manually; (ii) hand-write streaming AXI-Stream IP cores for the LBP, Sobel-gradient and per-cell color-mean front-ends on top of a 3-row line buffer; (iii) synthesise the full pipeline on PYNQ-Z2 with Vivado and measure end-to-end latency, targeting ≤ 100 μs per crop; (iv) tighten precision via hard-negative mining; (v) extend the disease vocabulary to include pebrine, flacherie and muscardine; and (vi) close the data loop, feeding low-confidence PYNQ inferences back to the Roboflow active-learning queue.

VI. REFERENCES

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